

# CS & IT ENGINEERING



## Computer Network - 1

### Transport Layer

Lecture No. – 08

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# Recap of Previous Lecture



Topic

TCP Flow Control





# Topics to be Covered



Topic

TCP Congestion Control

Topic

MSL



# ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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## Topic : Maximum Segment Size

→ Maximum Segment Size (MSS)

→ Maximum TCP segment payload size (in bytes)

$$\text{MSS} = \text{MTU} - \text{TCP Header Size} - \text{IP Header Size}$$

→ MSS clamping : MSS is specified during TCP handshake via SYN packet

MTU  $\Rightarrow$  Source Network MTU

Ex:- DLT  $\rightarrow$  Ethernet  
MTU = 1500 byte

$\Rightarrow$  IPv4 Base Header = 20 byte

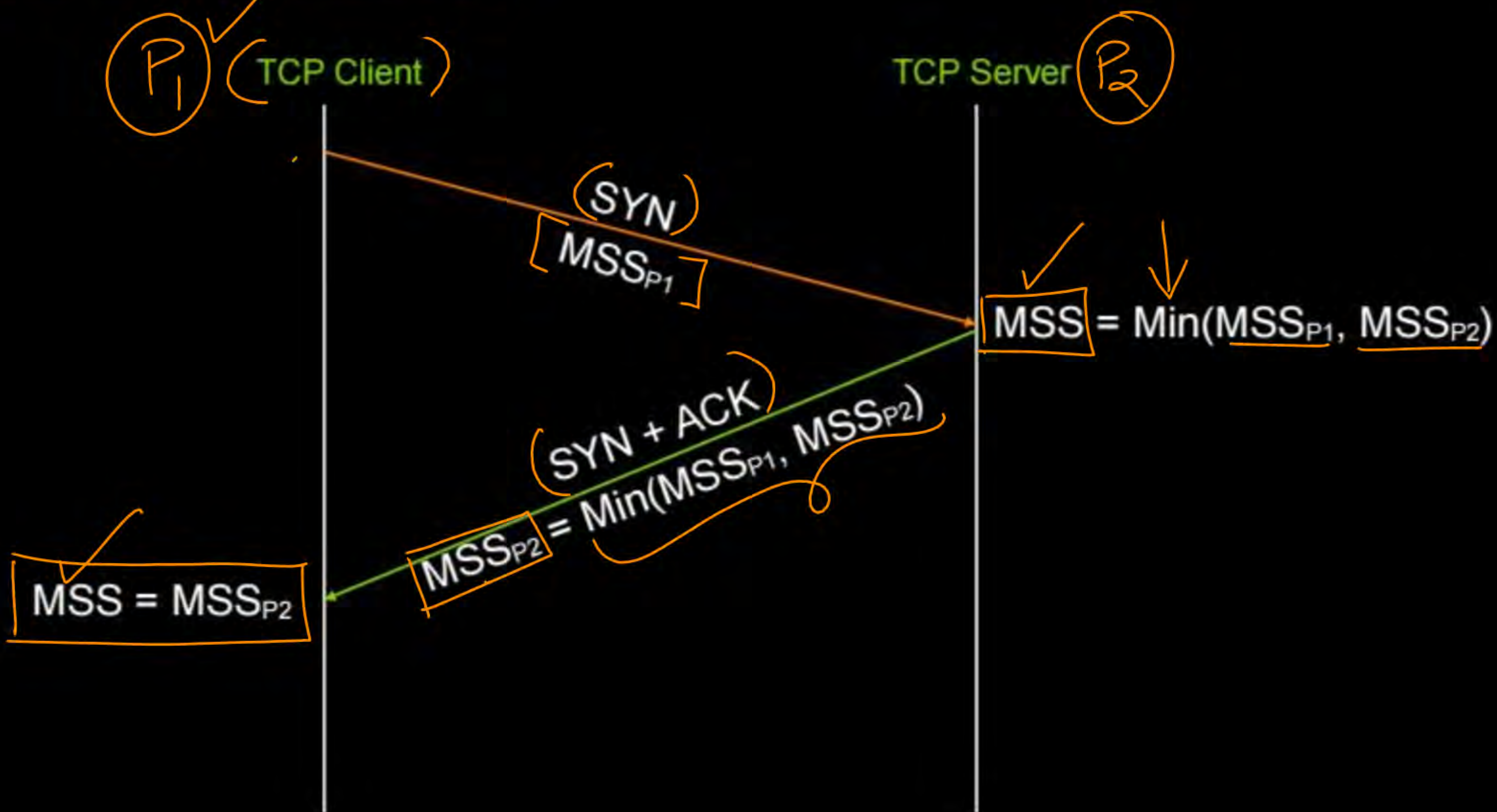
$\Rightarrow$  TCP Base Header = 20 byte

MSS = ?

$$\begin{aligned} \text{MSS} &= \left[ (\text{MTU} - \text{IPv4 Header Size}) - \text{TCP Header Size} \right] \\ &= \left[ (1500 - 20) - 20 \right] \text{ byte} \\ &= 1460 \text{ bytes} \end{aligned}$$



# Topic : MSS Clamping



TCP Client Process

$P_1$

SOCK\_STREAM  
(TCP Socket)

Send Buffer

(MaxSendBuffer = 1000 Bytes)

Recv Buffer

(MaxRecvBuffer = 1000 Bytes)

LastByteWritten =  $P_1$ 's ISN

LastByteSent =  $P_1$ 's ISN

LastByteAcked =  $P_1$ 's ISN

RWND: For Flow Control  
CWND: For Congestion Control  
Dynamic  
[Congestion Control Algo.]



LastByteRead =  $P_2$ 's ISN

NextByteExpected =  $P_2$ 's ISN + 1

LastByteRcvd =  $P_2$ 's ISN

TCP Server Process

$P_2$

SOCK\_STREAM  
(TCP Connection  
Socket 1)

Send Buffer

(MaxSendBuffer = 1200 Bytes)

Recv Buffer

(MaxRecvBuffer = 1200 Bytes)

LastByteWritten =  $P_2$ 's ISN

LastByteSent =  $P_2$ 's ISN

LastByteAcked =  $P_2$ 's ISN

RWND  
CWND

LastByteRead =  $P_1$ 's ISN

NextByteExpected =  $P_1$ 's ISN + 1

LastByteRcvd =  $P_1$ 's ISN



## Topic : Congestion Window

→ TCP maintain one more variable regarding 'Send Buffer', to control the congestion

→ Congestion Window Size (CWND)

$$[\text{LastByteSent} - \text{LastByteAcked}] \leq \text{minimum}[\text{CWND}, \text{RWND}]$$

No. of outstanding bytes

~~(Flow OR Congestion)~~  
(Flow AND Congestion)

CWND : Sender's Congestion Window Size

RWND : Receiver's advertised available space in Receiving Window



## Topic : Congestion Window

code → implemented



=> TCP wants to send a segment contains N bytes in payload

if [ ( LastByteSent - LastByteAcked ) + N ] ≤ minimum[ Cwnd, RWND ]

→ Fill upto N bytes [ (LastByteSent + 1) to (LastByteSent + N) ]  
in the payload field of segment from 'Send Buffer'

→ Sequence Number = LastByteSent + 1;

→ ACK Number = NextByteExpected;

→ LastByteSent = LastByteSent + N;

else

→ Not allowed to send immediately, wait for sometime.

## [MCQ]

11T-B, (H.W.) [GATE-IT-2005] [2 Mark] 

#Q. On a TCP connection, current congestion window size is Congestion Window = 4 KB. The window size advertised by the receiver is Advertise Window = 6 KB. The last byte sent by the sender is LastByteSent = 10240 and the last byte acknowledged by the receiver is LastByteAcked = 8192. The current window size at the sender is \_\_\_\_\_.

- ☐ A 2048 bytes
- ☐ B 4096 bytes
- ☐ C 6144 bytes
- ☐ D 8192 bytes



# Topic : Congestion Control



→ Two phase of TCP congestion control :

1. Slow Start Phase

[when  $CWND < SSThresh$ ]

2. Congestion Avoidance Phase

[when  $CWND \geq SSThresh$ ]

Algo.

Variable  
 $SSThresh$

update

RFC

$CWND$   
 $SSThresh$

Actually  
(in bytes)

MSS

int

For algo.  
(in MSS)

$SSThresh$  : Slow Start Threshold



## Topic : Slow Start Phase

→ When Congestion Window Size is less than Slow Start Threshold  
 $CWND < SSThresh$

→ Congestion Window ( $CWND$ ) grow exponentially ✖✖

→  $CWND$  increase by 1 MSS on every successful acknowledgment ✖✖

→  $CWND$  approximately doubles on every round trip time (cycle time) ✖✖✖

$$CWND = CWND + 1 MSS$$

SSThresh = [Infinite]

CWND = 1 MSS

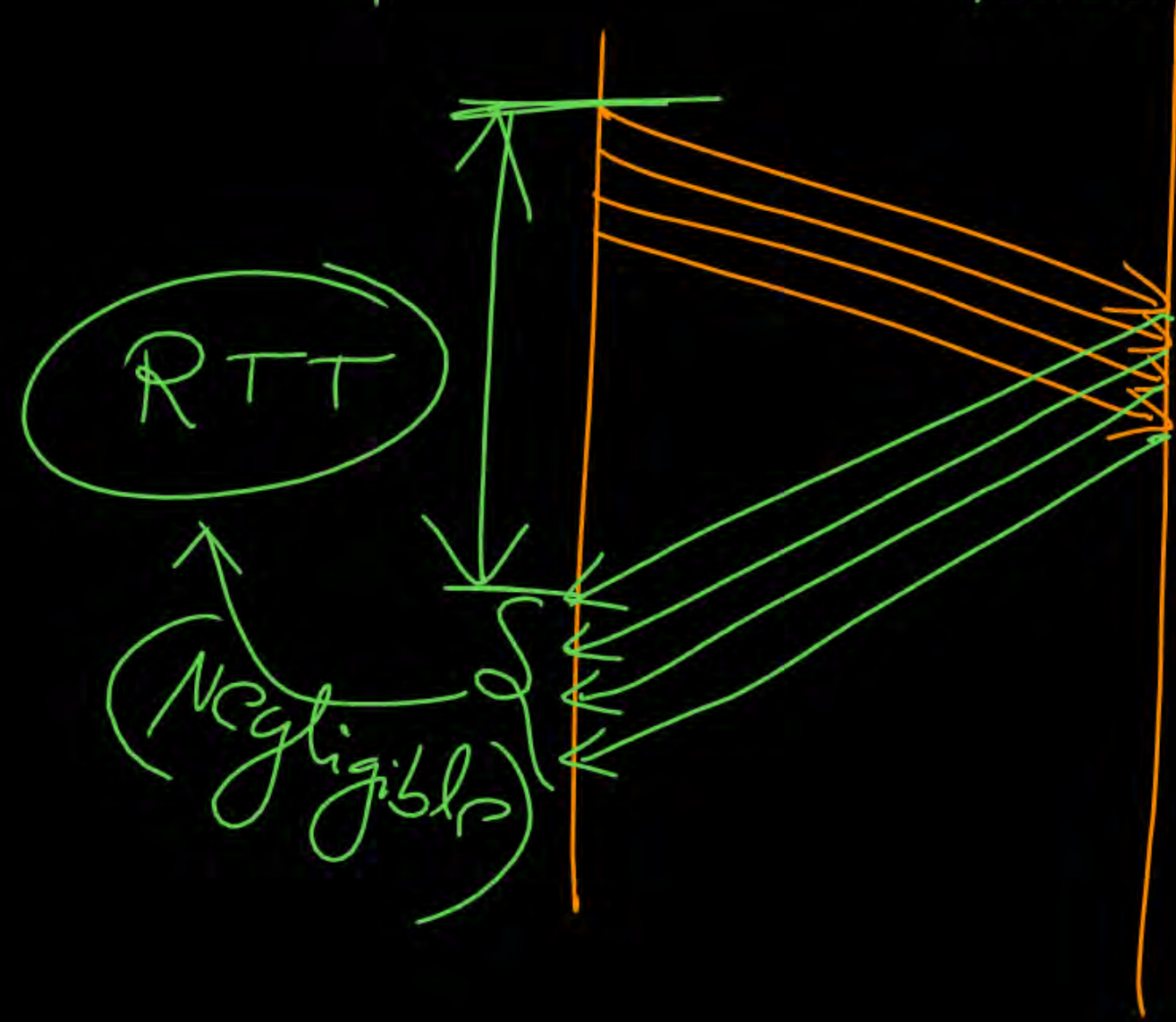
Exponen-  
-tial  
growth

2MSS

4MSS

8MSS

16MSS



[MCQ]

1156

[GATE-CS-2008] [2 Mark]



#Q. In the slow start phase of the TCP congestion control algorithm, the size of the congestion window: [CWND]

- ☒ A does not increase
- ☒ B increases linearly
- ☒ C increases quadratically
- ☒ D increases exponentially

Ans: D

[MCQ]

IIT-G

[GATE-2018] [1 Mark]



#Q. Consider the following statements regarding the slow start phase of the TCP congestion control algorithm. Note that cwnd stands for the TCP congestion window and MSS denotes the Maximum Segment Size.

- i. The cwnd increase by 2 MSS on every successful acknowledgement. FALSE
- ii. The cwnd approximately doubles on every successful acknowledgement. FALSE
- iii. The cwnd increase by 1 MSS every round trip time. FALSE
- iv. The cwnd approximately doubles every round trip time. TRUE

Which one of the following is correct?

- A Only (ii) and (iii) are true
- B Only (i) and (iii) are true
- ☒ C Only (iv) is true
- D Only (i) and (iv) are true

(i)  $\rightarrow$  T/F

Ans: C



## Topic : Congestion Avoidance Phase

→ When Congestion Window Size is greater then equal to Slow Start Threshold  
 $[CWND \geq SSThresh]$

→ Congestion Window (CWND) grow linearly \*\*

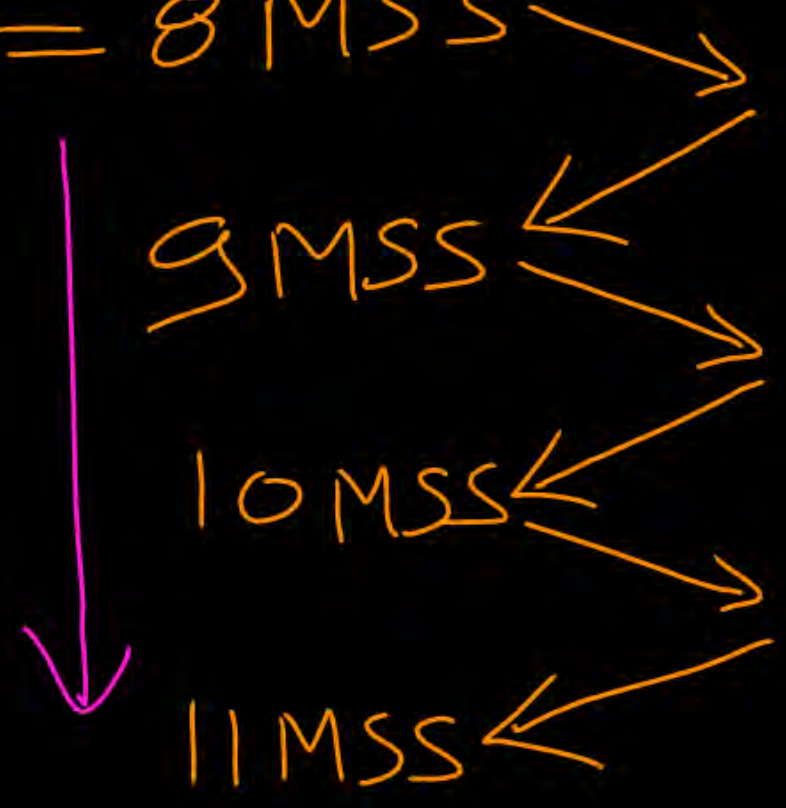
→ CWND increase by 1 MSS on every round trip time  
[On successful acknowledgments of all transmitted segments] \*\*

$$CWND = CWND + \frac{1}{[CWND]}$$

$SSThresh = 8 MSS$

$CWND = 8 MSS$

linear  
growth





## Topic : Congestion Control

→ When TCP connection established :  $P_1 \longleftrightarrow P_2$

Initialize :  $\text{SSThresh} = \text{Infinite} = (2^{16} - 1) * 2^{14}$   
 $\text{CWND} = [1 \text{ MSS}]$  ( $2 \text{ MSS} \rightarrow 10 \text{ MSS}$ )

→ When Time-out occur :

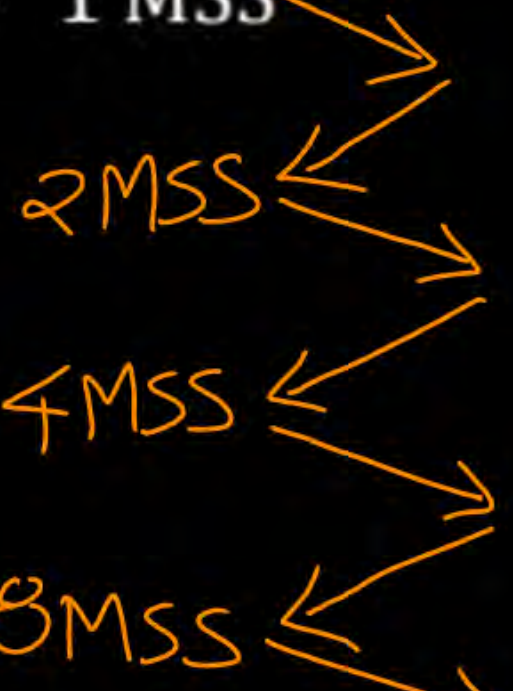
Update :  $\text{SSThresh} = [\text{CWND} / 2]$   
 $\text{CWND} = [1 \text{ MSS}]$

call Upto Programmer  
Floor



SSThresh = Infinite

CWND = 1 MSS

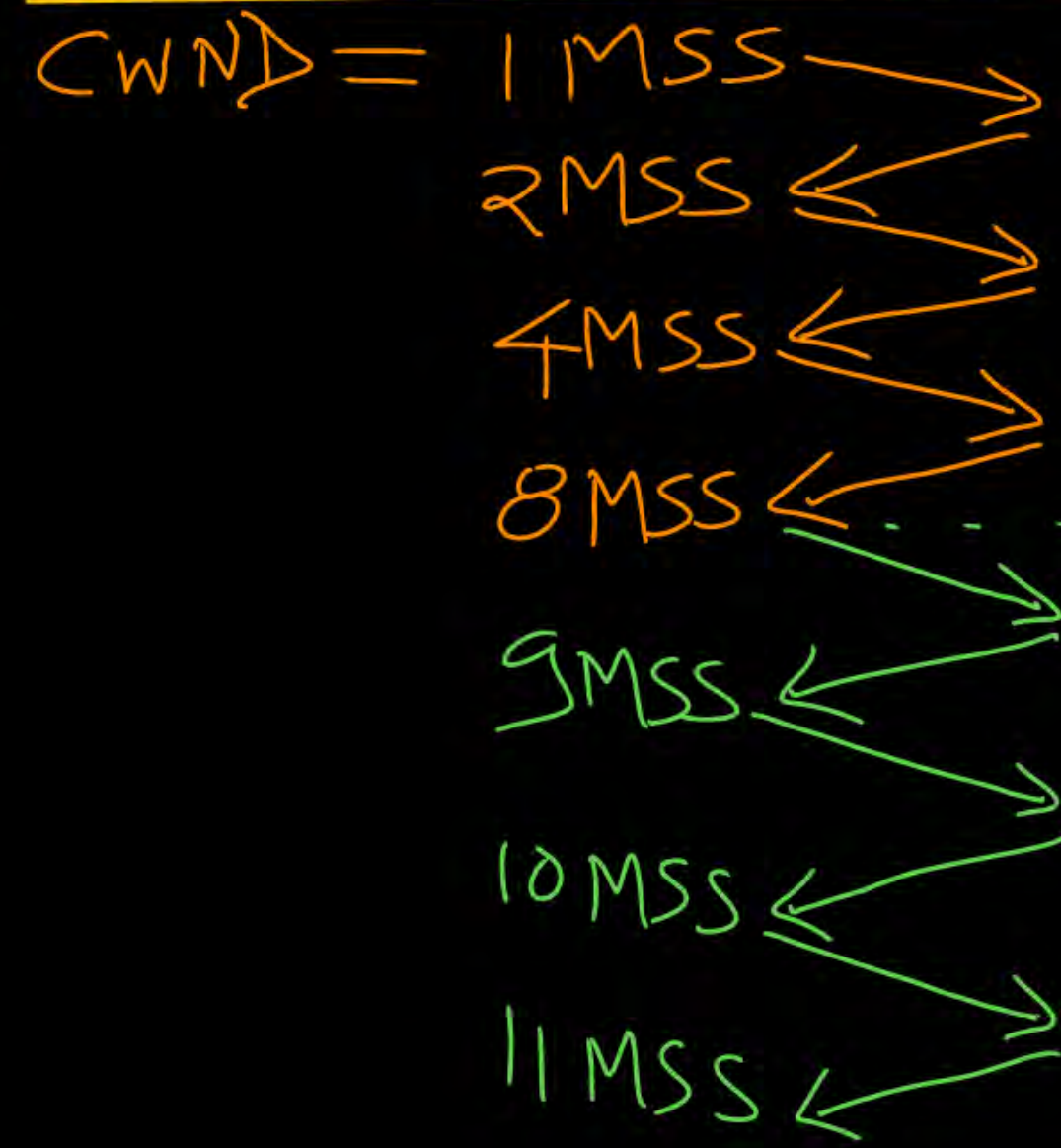


slow  
start  
phase

CWND = 16 MSS

Time-out

$$SSThresh = \frac{CWND}{2} = \frac{16MSS}{2}$$
$$SSThresh = 8MSS$$



slow  
start

Cong.  
Avoidance

[NAT]

115C

[GATE-2024][2 Mark]



#Q. Consider a TCP connection operating at a point of time with the congestion window of size 12 MSS (Maximum Segment Size), when a timeout occurs due to packet loss. Assuming that all the segments transmitted in the next two RTTs (Round Trip Time) are acknowledged correctly, the congestion window size (in MSS) during the third RTT will be \_\_\_\_\_.

$$CWND = 12 \text{ MSS}$$

Timeout

$$ssthresh = \frac{CWND}{2} = \frac{12 \text{ MSS}}{2} = 6 \text{ MSS}$$

$$CWND = 1 \text{ MSS}$$

$$2 \text{ MSS}$$

$$4 \text{ MSS}$$

1<sup>st</sup> RTT

2<sup>nd</sup> RTT

3<sup>rd</sup> RTT

Ans = 4

[NAT]

IIT-KGP, H.W.

[GATE-2014][2 Mark]



#Q. Let the size of congestion window of a TCP connection be 32 KB when a timeout occurs. The round trip time of the connection is 100 msec and the maximum segment size used is 2 KB. The time taken (in msec) by the TCP connection to get back to 32 KB congestion window is \_\_\_\_\_.

[NAT]

IIT-D, H.W.

[GATE-2020][2 Mark]



#Q. Consider a TCP connection between a client and a server with the following specifications; the round trip time is 6 ms, the size of the receiver advertised window is 50 KB, slow-start threshold at the client is 32 KB, and the maximum segment size is 2 KB. The connection is established at time  $t=0$ . Assume that there are no timeouts and errors during transmission. Then the size of the congestion window (in KB) at time  $t+60$  ms after all acknowledgments are processed is \_\_\_\_\_ .

[MCQ]

IIT-D, (H.W.)

[GATE-2012][2 Mark]



#Q. Consider an instance of TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm where the window size at the start of the slow start phase is 2 MSS and the threshold at the start of the first transmission is 8 MSS. Assume that a timeout occurs during the fifth transmission. Find the congestion window size at the end of the tenth transmission.

- ☐ A 7 MSS
- ☐ B 8 MSS
- ☐ C 12 MSS
- ☐ D 14 MSS

(IIT-D Anskey)  
Marks to All

[MCQ]

IIT-D, H.W.

[GATE-IT-2004] [2 Mark]



#Q. Suppose that the maximum transmit window size for a TCP connection is 12000 bytes. Each packet consists of 2000 bytes. At some point of time, the connection is in slow-start phase with a current transmit window of 4000 bytes. Subsequently, the transmitter receives two acknowledgments. Assume that no packets are lost and there are no time-outs. What is the maximum possible value of the current transmit window ?

- ☐ A 4000 bytes
- ☐ B 8000 bytes
- ☐ C 10000 bytes
- ☐ D 12000 bytes



## 2 mins Summary



Topic

TCP Congestion Control

(V.V.I m P)

Topic

~~MSL~~



# THANK - YOU

